

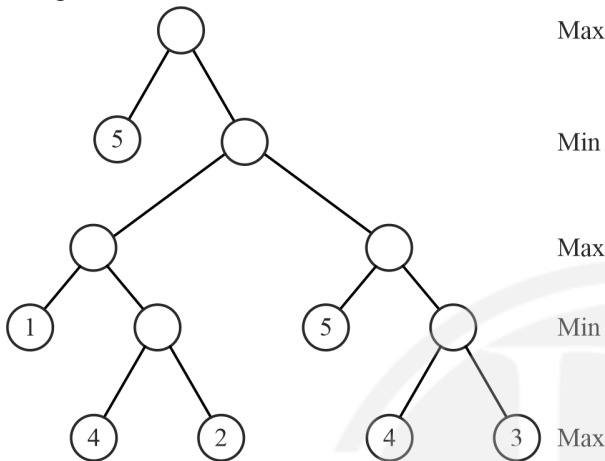
DS&AI

Artificial Intelligence

Adversarial search

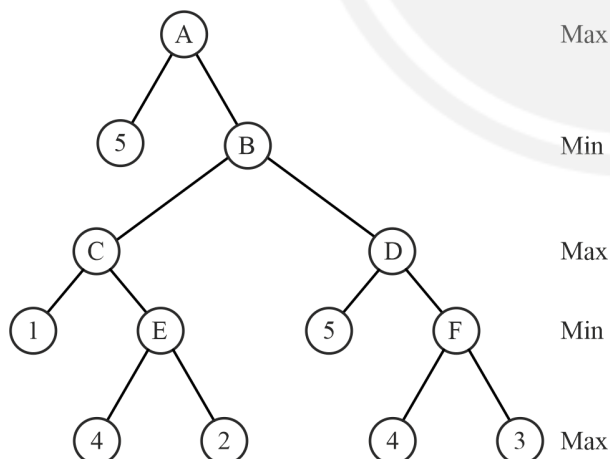
DPP 02

Q1 What is the minimax value of the root node for the game tree below?



Assuming that it performs a depth-first search that always generates the leftmost child node first and a loss (and win) of MAX (and MIN) corresponds to a value of $-\infty$ (and ∞ , respectively).

Q2 Consider the below search tree



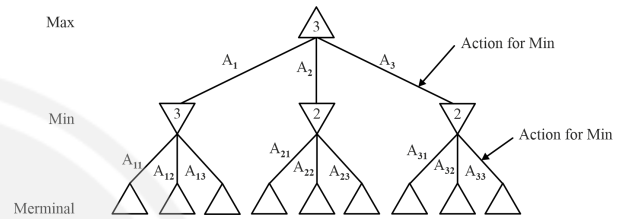
If we apply α - β pruning method to above tree, then which of the node(s) will get pruned?

Assuming that it performs a depth-first search that always generates the leftmost child node first and a loss (and win) of MAX (and MIN) corresponds to a value of $-\infty$ (and ∞ , respectively).

Q3 What is the time complexity of the Minimax algorithm in terms of the branching factor (b) and the depth of the search tree (d)?

- (A) $O(b^d)$ (B) $O(b.d)$
(C) $O(b^{d/2})$ (D) $O(b^b)$

Q4 Draw same diagram 5 also



Consider the following game tree where MAX and MIN take turns. The terminal node values are given at the bottom of the tree.

Assume both players play optimally.

- (i) What is the value at the root node?
- (ii) Which action should MAX take at the root node?

- (A) Value = 3, Action = A_1
(B) Value = 2, Action = A_2
(C) Value = 2, Action = A_3
(D) Value = 4, Action = A_2

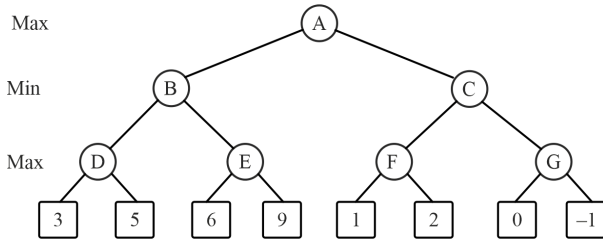
Q5 Which of the following is/are false regarding the alpha-beta pruning for the mini-max search algorithm?

- (A) It can potentially lead to suboptimal solutions compared to mini-max search without any pruning
- (B) It is guaranteed to improve the running time in-comparison to the mini-max search without any pruning
- (C) The order in which nodes are visited affects the amount of pruning
- (D) If the successors of a node are chosen randomly, the time complexity (on average) is



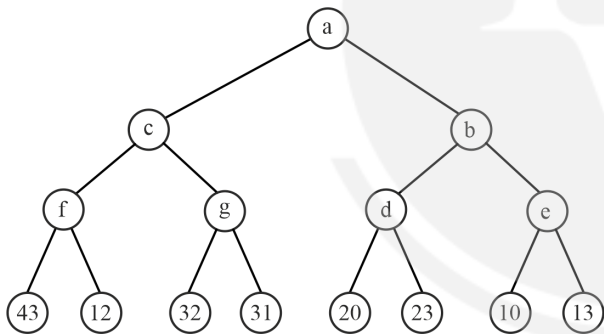
$$O(b^{3m/4})$$

- Q6** In the following two-ply game tree, the terminal nodes show the utility values computed by the utility function. Use the Minimax algorithm to compute the utility values for other nodes in the given game tree.

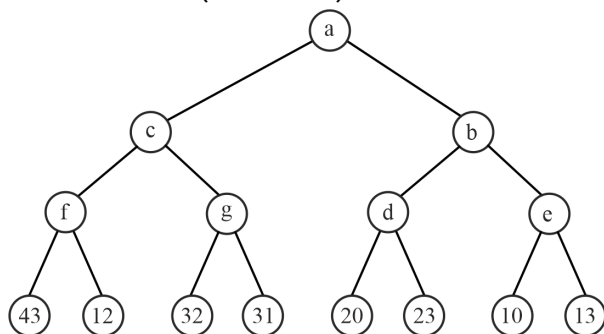


What will be the utility value for Node A, B and C?

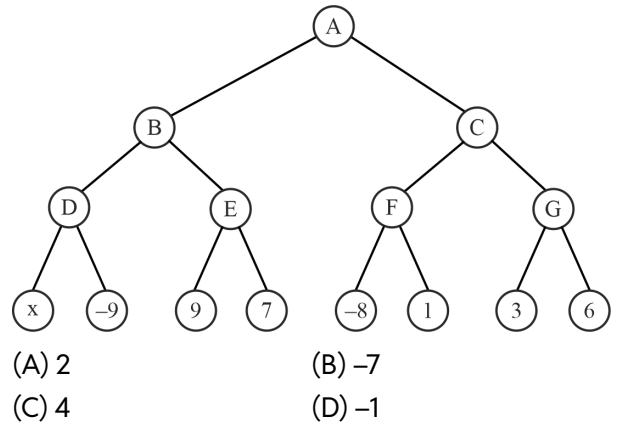
- (A) $A = 5, B = 5$ & $C = 0$
 (B) $A = 0, B = 5$ & $C = 0$
 (C) $A = 0, B = 9$ & $C = 2$
 (D) $A = -1, B = 3$ & $C = -1$
- Q7** What is the value of the root node after MiniMax search over the given tree? The root node (labelled 'a') is a max node.



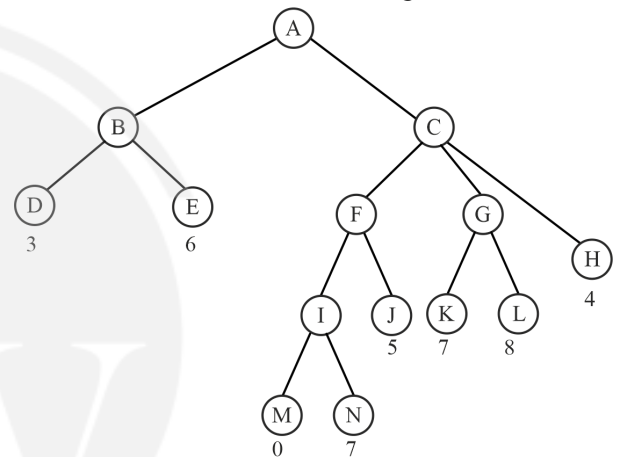
- Q8** How many nodes (including leaves and internal nodes) will an alpha-beta pruning based MiniMax search visit in the given tree, assuming each successor is ordered left to right?
 The root node (labelled 'a') is a max node.



- Q9** Player 1 (max node) chooses the left action for which of the following value(s) of x . Assume optimal play and Player 1's turn at Node A

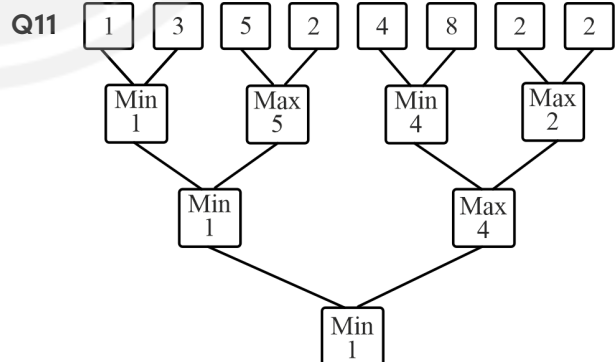


- Q10** Perform minmax on the following tree-



Explain each step.

Perform Minimax Search on the given game tree.



Consider the following game tree where MIN and MAX alternate turns. Terminal node values are given at the bottom level.

- Perform the Minimax algorithm to compute the value at the root node.

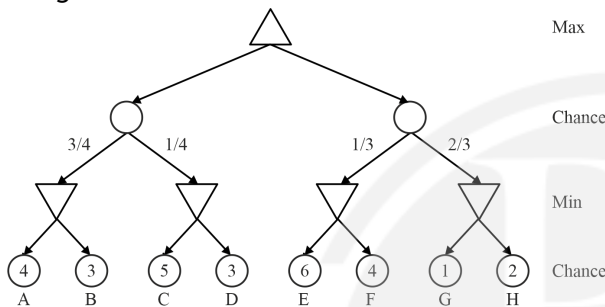


- Identify which branch the root node will choose.

(Hint: At MIN nodes, choose the minimum value; at MAX nodes, choose the maximum value.)

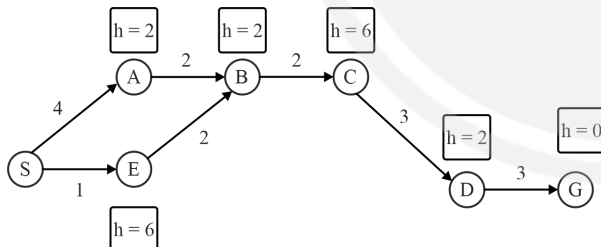
- (A) Root value = 4, Branch = Right Subtree
 (B) Root value = 1, Branch = Left Subtree
 (C) Root value = 2, Branch = Middle Subtree
 (D) Root value = 5, Branch = Left Subtree

- Q12** Consider the following two-player game tree in which the game has an element of chance, which is shown by the so-called probability nodes in the game tree.



What is the correct value for the top MAX node if one applies the expect minimax algorithm?

- Q13** The state graph shows the action cost along the edges and the heuristic function h associated with each state.



Suppose A^* algorithm is applied on this state graph using priority queue to store the frontier. In what sequence are the nodes expanded?

- (A) S, A, E, C, B, D, G
 (B) S, E, A, C, B, D, G
 (C) S, A, E, B, C, D, G
 (D) S, A, B, E, C, D, G

- Q14** For more complex games, such as chess or checker the AND/OR search to termination is out of question. Which of the following statements are true?

- Our goal in searching such a game tree might be, instead, merely to find a good first move.
- Extract from the search graph an estimate of the 'best' first move

- (A) Both 1 and 2
 (B) Only 2
 (C) Only 1
 (D) Both 1 and 2 are false

- Q15** Using alpha-beta pruning, in the best case, you examine only $(2b)^{d/2}$ leaf nodes. The result is you can search _____ as deep as minimax!
- (A) Equal (B) Twice
 (C) Half (D) One-fourth



Answer Key

Q1 0~0

Q2 5~5

Q3 (A)

Q4 (A)

Q5 (C)

Q6 (A)

Q7 0~0

Q8 0~0

Q9 (A)

Q10 4~4

Q11 (B)

Q12 3~3

Q13 (C)

Q14 (A)

Q15 (B)



[Android App](#)



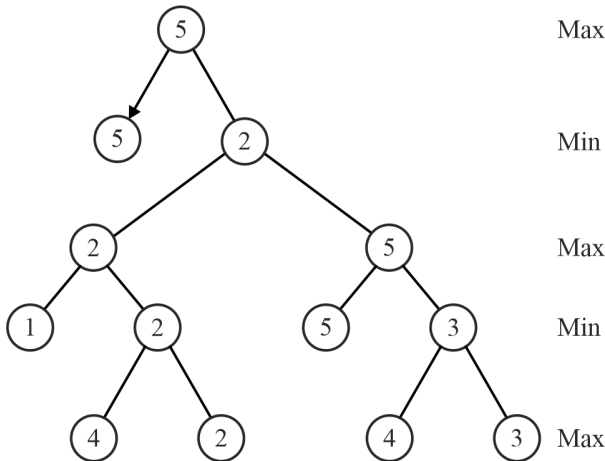
[iOS App](#)



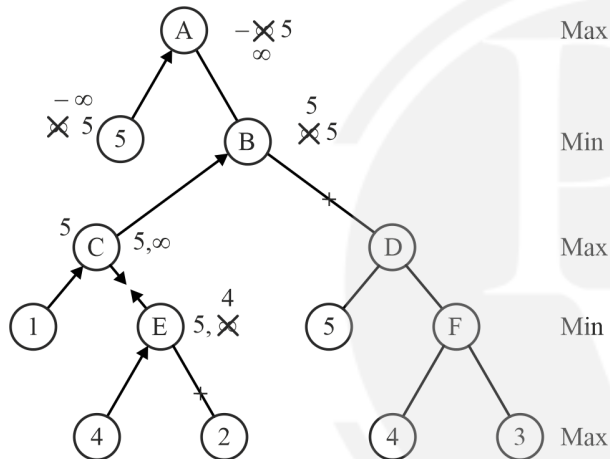
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Hints & Solutions

Q1 Text Solution:



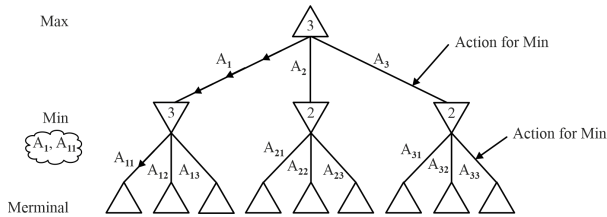
Q2 Text Solution:



Q3 Text Solution:

$O(b^d)$

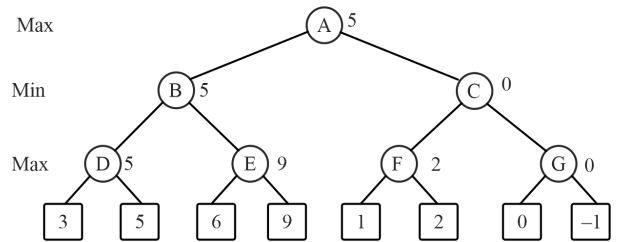
Q4 Text Solution:



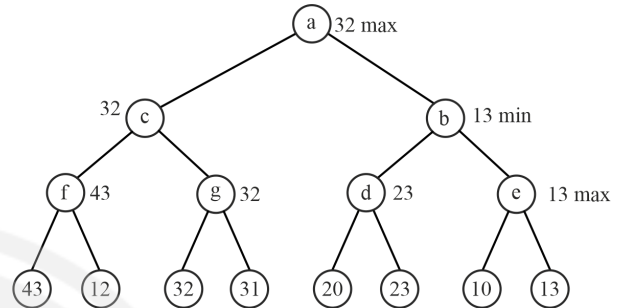
Q5 Text Solution:

- The order in which nodes are visited affects the amount of pruning

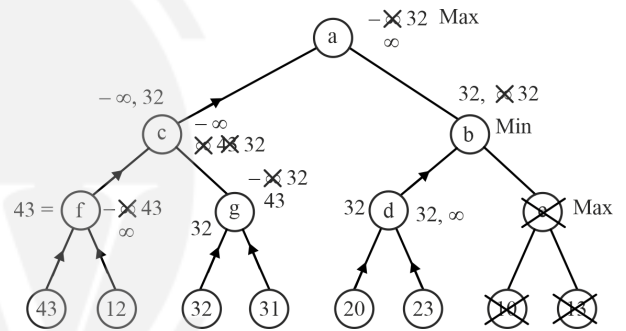
Q6 Text Solution:



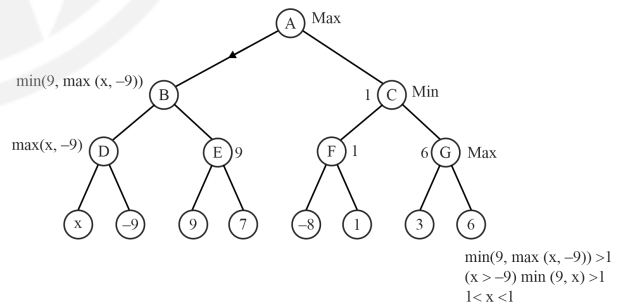
Q7 Text Solution:



Q8 Text Solution:



Q9 Text Solution:

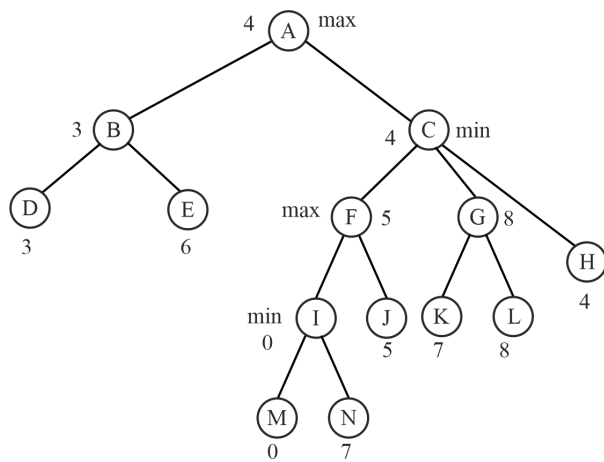


$\min(9, \max(x, -9)) > 1$
 if $x > -9$, $\min(9, x) > 1$
 $1 < x < 9$, true
 $x > 9$, $9 > 1$ true
 $x > 1$

Q10 Text Solution:



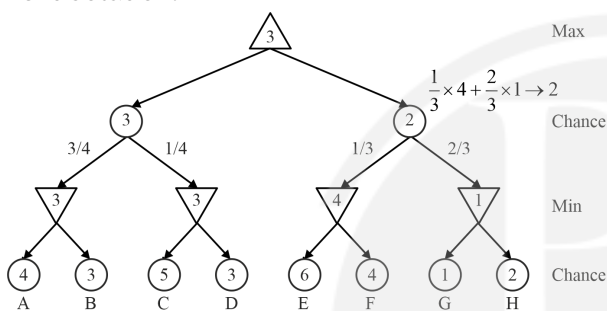
Android App | iOS App | PW Website



Q11 Text Solution:

- Root value = 1, Branch = Left Subtree

Q12 Text Solution:



$$\sum P \times \text{value}$$

$$\frac{3}{4} \times 3 + \frac{1}{4} \times 3 \rightarrow 3$$

Q13 Text Solution:

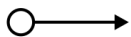
S

A 6

ϵ 7

B 8 5

Q14 Text Solution:



Start end

We just want to find best move based on current situation of game

Q15 Text Solution:

Minimax b^d

α - β Minimax $b^{d/2}$

